CARDIOTWIN

AI-Powered Cardiac & CAD Health Companion

Predict · Explain · Guide · Connect

- **Problem Statement -** To address the lack of early cardiac risk awareness, explainable predictions, personalized health guidance, and accessible emergency support.
- **Theme –** MedTech / BioTech / HealthTech
- **PS Category –** Software
- **Team Name –** AyuRaksha



CARDIOTWIN: THE IDEA

Problems

- Late detection of cardiac deterioration
- Scattered EHR, lab & wearable data
- Black-box AI with no explanations
- Generic, one-size-fits-all thresholds
- Missing or stale sensor readings
- No "what-if" view of risk changes
- Limited real-time access to alerts

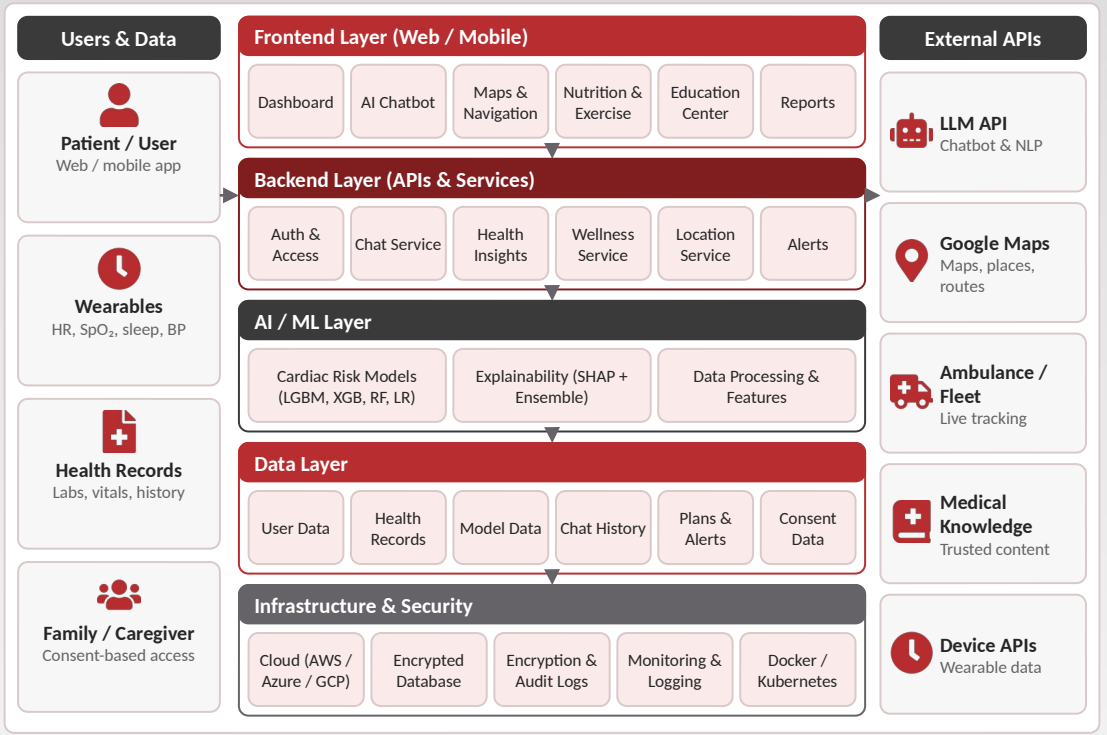
Proposed Solution

1. **Data Integration:** Combines EHR, lab results and wearable vitals.
2. **Dual AI Models:** Full-data and watch-only cardiac risk prediction.
3. **Explainable AI:** SHAP insights with SAR doctor reports.
4. **Personalized Risk Engine:** Patient baselines with Low / Moderate / High tiers.
5. **Emergency Alerts:** Independent warnings with a user-confirmed 112 call.
6. **What-If Simulator:** Predicts risk changes under different health scenarios.
7. **Smart Delivery:** Android app, virtual watch and Wear OS (planned).

Proof of concept built on public / replayed clinical data — not a diagnostic device.

Personalized Risk Tiers

LOW	MODERATE	HIGH
Within personal baseline	Score ≥ 3.62% · watch closely	Score ≥ 7.59% · alert + report



Innovation & Uniqueness

- **Personalized Monitoring:** patient-specific baselines, not generic thresholds.
- **Dual AI Models:** watch-only and full-data risk prediction.
- **Safety-First Design:** stale-data checks and independent emergency alerts.
- **Explainable AI:** SHAP insights with doctor-friendly SAR reports.
- **Smart What-If Simulation:** counterfactuals and stability checks.
- **Reliable Predictions:** input validation and transparent limitations.
- **Flexible Integration:** one backend for mobile, smartwatch and ESP32.

TECHNICAL APPROACH

OFFLINE · BUILD AND VALIDATE (NOTEBOOKS 01–06)

INPUT
Data sources
MIMIC-IV clinical demo
100 patients · 140 ICU stays
MIMIC-IV-ECG: 659 12-lead
Waveform DB: 19 usable
I-CARE: metadata only

→

NOTEBOOK 01
Structure
11 tidy CSV tables
12,101 ICU-hours of vitals
34,587 labs · 2,856 meds
288 labelled events
ECG features, 10 s @ 500 Hz

→

NOTEBOOK 02
Merge and clean
static · clinical · dynamic
range limits, °F → °C fix
forward-fill ≤ 4 h per stay
NEWS2 + shock index
shared twin_preprocess.py

→

NOTEBOOKS 03–04
Features + EDA
+65 cardiac features → 211
AF burden, HF warning signs
SpO_2 dips, QTc, ST, LVH
meds, labs, ICD risk factors
11 EDA charts incl. shift

→

NOTEBOOK 05
Train and select
deterioration within 24 h
59 train (5 folds) / 15 test
LR · RF · XGB · LGBM
watch: LR, 46 inputs
full: LightGBM, 40 inputs

→

NOTEBOOK 06
Explain
exact SHAP per prediction
19 patient waterfalls
19 doctor reports
stability + subgroups
watch model: 50% / 50%

Saved, versioned artifacts: model .pkl + SHA-256 manifest · feature schema · cut-offs MODERATE ≥ 3.62%, HIGH ≥ 7.59% · 50 bootstrap refits · SHAP background

ONLINE · EVERY HOUR, PER PATIENT (NOTEBOOK 07)

LIVE INPUT
Readings
Watch: heart rate, SpO_2, breathing rate, rhythm
Phone: age, sex, smoking, symptom check
Records: labs, history
Tested with ICU vitals replayed as watch data

→

RISK ENGINE · 48 TESTS PASS · 99.7% SAME TIER AS NOTEBOOK 05

1 Validate units, duplicates, impossible values →	**2 Emergency rules** NEWS2 red bands, chest pain →	**3 Freshness gate** stale signal: no score, never LOW →	**4 Features** same code as training →	**5 Model** watch LR, SHA-256 integrity check
6 Policy LOW · MOD · HIGH with hysteresis →	**7 Uncertainty** 50 bootstrap refits, 24 h trajectory →	**8 SHAP** exact linear SHAP, top 3 reasons →	**9 SAR narrative** every sentence carries its evidence →	**10 Audit + alerts** de-duplication, cooldown, audit log

Emergency rules run before the model and independently of it; a triggered rule overrides any score.

→

OUTPUT
What people see
Patient phone: tier in words, no score
Doctor phone: score, reasons, SAR, audit
Watch: one word + haptic
Emergency: Call 112, user presses it

NOTEBOOK 08 · WHAT-IF SIMULATOR: 21 levers · sensitivity curves · counterfactuals · reuses saved models, nothing retrained · labelled HYPOTHETICAL

Planned — Application & Deployment

Data pipeline and ML components are built; backend, mobile and smartwatch integration are planned.

- **Backend** — FastAPI · SQLite / PostgreSQL · REST · WebSocket
- **Simulator** — Python Virtual Watch
- **Mobile App** — Android · Kotlin · Jetpack Compose
- **Smartwatch** — Wear OS · Health Services · Bluetooth
- **Deployment** — Render / Railway

Tech Stack (Built)

- **Language** — Python, Jupyter Notebook
- **Data Processing** — Pandas, NumPy
- **ML Models** — LightGBM, XGBoost, Random Forest, Logistic Regression
- **Optimization** — Optuna, Scikit-learn
- **Explainable AI** — SHAP, Bootstrap Ensemble
- **Risk Engine** — Python, YAML, Pytest
- **Visualization & Reports** — Matplotlib, Markdown, HTML

FEASIBILITY AND VIABILITY

Challenges / Risks & Strategies

Challenge / Risk	Strategy
1. Missing / stale sensor data	Freshness gate: stale input never scores LOW
2. Black-box predictions	Exact SHAP + evidence-backed SAR report
3. Missed emergencies	Emergency rules run before & independent of model
4. Noisy or impossible readings	Validation of units, duplicates and ranges
5. Unstable risk scores	50 bootstrap refits + tier hysteresis
6. Alert fatigue	De-duplication, cooldown and audit log
7. Clinical & regulatory risk	Not diagnostic; clinical validation before use

Key Performance Indicators

211 engineered cardiac features

48 risk-engine tests pass

99.7% same tier as validated model

21 what-if levers

Feasibility of the Idea

Technical — High

Extends the existing ML pipeline with chatbot APIs, frontend components, backend endpoints and Google Maps; reusing current models cuts development effort.

Economic — Moderate to High

Incremental build is cheaper than a rebuild; AI-API, Maps, hosting, security and live ambulance integrations add recurring costs.

Operational — High (software)

Chatbot, CAD education, wellness tips and reports fit the current UI; ambulance dispatch and live tracking need external coordination.

Data — Moderate

Risk scores and SHAP explanations are reusable; insights depend on data completeness, timestamps, sensor accuracy and CAD validation.

Safety & Regulatory — Requires validation

Educational features are achievable now; CAD risk, clinical advice and emergency detection need clinical validation, privacy safeguards and regulatory review.

Viability

Healthcare

High potential

Helps users understand risk factors, spot warning signs, track trends and prepare for doctor visits, within clear clinical boundaries.

User Adoption

High potential

A conversational interface, simple explanations, Hindi support and accessible controls are easier than complex dashboards.

Business

Conditional

Individuals, clinics, cardiac-care providers and health programs; depends on proven usefulness, trust, partnerships and willingness to pay.

Long-term Sustainability

Moderate to high

Modular design adds features incrementally; budget for model monitoring, API costs, content updates, security and wearable changes.

IMPACT AND BENEFITS

Potential Impact

- **Heart Health Awareness** — educates the public on CAD and heart-disease prevention.
- **Early Risk Awareness** — helps identify potential cardiac risk factors early.
- **Healthy Lifestyle** — encourages better nutrition and physical activity.
- **Easy Access** — convenient, AI-based health information in plain language.
- **Emergency Support** — links cardiac alerts with hospital navigation.
- **Wider Reach** — cardiac health education for underserved communities.

Benefits

Improved cardiac health awareness
Understand CAD, risk factors, symptoms and prevention through accessible explanations.

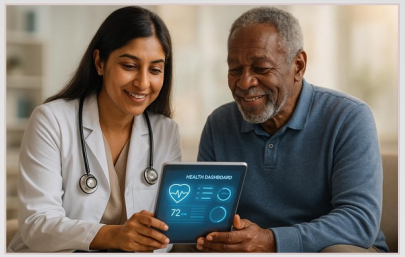
Easier access to health information
Ask questions in natural language instead of navigating complex medical dashboards.

Better emergency accessibility
Emergency contacts, hospital locations and supported ambulance tracking in one alert flow.

Healthier lifestyle decisions
Personalized nutrition, exercise guidance and daily wellness goals suited to each user.

More understandable AI predictions
SHAP shows which factors drive the risk score, supporting informed talks with doctors.

Benefits in Action



Heart health awareness



AI chatbot access



Emergency support



Healthy Indian diet



Explainable AI for doctors

Better Heart Health for a Healthier Tomorrow

Who Benefits

Patients & Families

- Risk tier in plain words
- Daily wellness goals
- User-confirmed 112 call

Doctors & Clinics

- Score + top SHAP reasons
- SAR report and audit log
- Better-prepared consultations

Public Health

- CAD awareness at scale
- Hindi-language support
- Reach underserved communities

RESEARCH AND REFERENCES

Research Work

01 Cardiac Risk Prediction

ML models estimate cardiac risk from available health data.

02 Explainable AI

SHAP explains the factors influencing model predictions.

03 CAD Health Education

Evidence-based information about coronary artery disease, risk factors and prevention.

04 Emergency Cardiac Support

Integrates the existing emergency alert system and hospital navigation.

References & Data Sources

UCI Machine Learning Repository
Public datasets for research and benchmarking.

scikit-learn Documentation
Model training, evaluation and calibration.

SHAP Documentation
Explainable machine learning.

American Heart Association
Cardiovascular health education and prevention guidance.

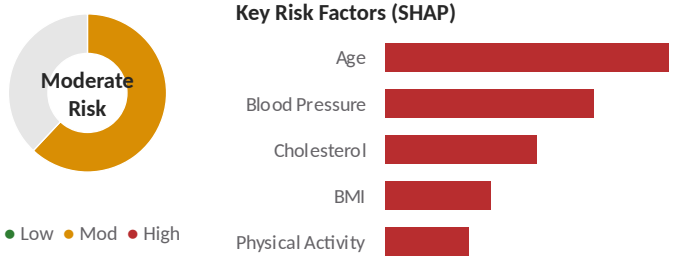
Comparison with Existing Features

Feature	Basic Risk Calculator	General Health App	CardioTwin
Cardiac risk estimation	✓	◐	✓
Explainable predictions	×	×	✓
CAD education	×	◐	✓
Lifestyle guidance	×	◐	✓
Emergency alert integration	×	◐	✓
Hospital map navigation	×	◐	✓

✓ Supported ◐ Partial / Conditional × Not supported

Project Insights

1. Cardiac Risk Dashboard & Explainability



2. AI Chatbot – CAD Education

What is Coronary Artery Disease (CAD)? Can you tell me about its risk factors?

CAD occurs when arteries supplying the heart narrow due to plaque build-up. Common risk factors include high blood pressure, high cholesterol, diabetes, smoking, obesity, inactivity and family history.

Type your message...

3. Nearby Hospitals & Ambulance Map

City Hospital 2.3 km

Sunrise Medical Center 4.1 km

Care Hospital 5.8 km

ETA
8 min

Research prototype — clinical validation required; not a diagnostic device.